G. Alarm List and Alarm Documentation

Appendix G of the EFC Gases & Advanced Materials Controls Standard, Revision I, October 14, 2026. Standalone copy; see the full document for the other chapters and appendices.

[Future: the alarm documentation described in this appendix is not implemented at EFC yet. It shows what a formal alarm-management practice would look like so it can be discussed and adopted in stages.]

G.1 The Standards

Alarm management is governed by ANSI/ISA-18.2 and its international equivalent IEC 62682. Both define an alarm *lifecycle*: write the rules (philosophy), decide which alarms are justified and how each should be set (rationalization), build them, operate and maintain them, and measure how the system performs. ISA has also published technical reports that explain each step in more depth.[1]

G.2 The Three Main Documents

- **Alarm philosophy** — the site rules: what qualifies as an alarm, how priorities are assigned, colors and sounds, shelving and suppression rules, who approves changes, and performance targets. Written once for the site and reused on every project.
- **Rationalization** — the review of each proposed alarm against the philosophy by a small team (process, operations, controls). Each alarm is either justified and fully defined, or rejected with a reason.
- **Master alarm database (alarm list)** — the result: one record per alarm with all of the fields below. It is the source for configuring Ignition and for testing.

G.3 Alarm Record Fields

Table G.1: Alarm Record Fields

Field	Meaning
Tag / alarm ID	Unique identifier, normally the instrument tag plus the alarm type (PT-210.HI).
Description	Short message shown to the operator.
Alarm type	High, low, deviation, rate of change, discrete (state), fault, or bad quality.

[1] ANSI/ISA-18.2-2016, *Management of Alarm Systems for the Process Industries*; IEC 62682:2022 (Ed. 2); ISA technical reports TR18.2.1 (alarm philosophy, 2018), TR18.2.2 (identification and rationalization, 2016), TR18.2.3 (basic alarm design), TR18.2.4 (enhanced and advanced alarm methods), TR18.2.5 (monitoring, assessment, and auditing), TR18.2.6 (batch and discrete processes), and TR18.2.7 (packaged systems).

Table G.1 continued

Field	Meaning
Setpoint	Value at which the alarm activates, in engineering units.
Deadband	How far the value must return before the alarm clears; prevents chattering.
On / off delay	How long the condition must persist before the alarm activates or clears; filters short spikes.
Cause	What most likely made the condition happen.
Consequence	What happens if the operator does nothing.
Operator response	The specific action the operator takes.
Maximum response time	How long the operator has before the consequence occurs.
Priority	Assigned from consequence and response time (Section G.5).
Operating-state applicability	Process states in which the alarm is meaningful (for example running but not shut down).
Suppression / shelving rules	When the alarm is suppressed by design, and who may shelve it and for how long.
Setpoint basis	Where the setpoint came from (PHA, equipment limit, operating experience).
Test method	How the alarm is proven during FAT and commissioning.

G.4 Worked Examples

Table G.2: Example Alarm List Entries

Tag	Type	SP / DB / delay	Priority	Cause → consequence	Response (time)
PT-210.HI	High process	50 torr / 5 torr / 10 s	Medium	Vacuum pump degraded or header leak → evacuation stalls	Check pump and header (15 min)
TT-305.LO	Low process	−180 °C / 2 °C / 30 s	High	Loss of cooling → column upset and off-spec product	Check refrigeration; reduce feed (5 min)
FIC-120.DEV	Deviation	±10% of setpoint / 2% / 60 s	Low	Valve sticking or loop out of tune → slow drift from target	Check valve position; switch to manual if needed (30 min)
M-101.FLT	Discrete fault	VFD fault bit / — / 0 s	Medium	Drive trip → pump stops, station cannot fill	Read VFD fault code; reset or call maintenance (15 min)
PT-210.BADQ	Bad quality	Signal <3.6 mA or comm loss / — / 5 s	Medium	Transmitter or wiring failure → loss of pressure indication and permissive	Treat value as unknown; check transmitter (10 min)
PT-210.LO	Low process (state suppressed)	5 torr / 1 torr / 10 s	Low	Only meaningful while venting; suppressed by design in other states	Check vent valve (30 min)

Table G.2 continued

Tag	Type	SP / DB / delay	Priority	Cause → consequence	Response (time)
LT-400.HI	*Rejected*	—	—	Duplicates the HH alarm on the same tank and has no separate operator action	Rejected in rationalization; HH alarm retained

G.5 Assigning Priority

Priority comes from two questions: how bad is the consequence, and how much time does the operator have? Equipment importance alone does not set priority.

Time to respond	Minor consequence	Major consequence	Severe consequence
More than 30 minutes	Not an alarm (use a warning)	Low	Low
10 to 30 minutes	Low	Low	Medium
3 to 10 minutes	Low	Medium	High
Less than 3 minutes	Medium	High	High

A typical healthy distribution is roughly 80% low, 15% medium, and 5% high priority alarms. Consequence categories (minor, major, severe) are defined in the site alarm philosophy for safety, environment, product, and equipment.

G.6 Performance Targets

The following benchmarks, commonly cited from ISA-18.2, are guidance for reviewing alarm performance after startup:

- Average alarm rate per operator: about 6 per hour is very likely acceptable; about 12 per hour is the most that is manageable.
- Alarm floods (more than 10 alarms in 10 minutes): less than 1% of the time.
- The 10 most frequent alarms: less than 1 to 5% of all alarms.
- Chattering and fleeting alarms: zero.
- Standing (stale) alarms: fewer than 5 on any day, with a plan to clear them.
- Unauthorized suppression: zero.

G.7 Where It Fits in Ignition

- Alarms are configured on tags, normally in the Ignition UDT for each equipment type (Section 10.4 of the Controls Standard), with setpoint, deadband, delays, priority, and the display message from the alarm list. Cause, consequence, and response can be stored as associated data and shown in the alarm faceplate.
- State-based suppression is driven by expressions on the alarm's enabled property.
- Shelving is limited by role and time in the alarm status table configuration.

- The alarm journal records activations, acknowledgments, and shelving for the performance review; notification pipelines handle call-outs.

See also Section 5.7 of the Controls Standard and the Alarms section of Chapter 11.